

Jad El Badaoui

Aerospace Engineer | Flight Physics: Aerodynamics, CFD and Wind Tunnel Testing

Bristol, United Kingdom | (+44) 7939 273309 | jadbadaoui00@gmail.com

jadelbadaoui.com | linkedin.com/in/jad-el-badaoui | github.com/Jadbadaoui

PROFILE

First Class BEng Aerospace Engineering graduate from the University of Bristol, top of the cohort in Aerodynamics (95%) and Engineering Mathematics (94%), progressing to Imperial College London's MSc in Advanced Computational Methods for Aeronautics, Flow Management and Fluid-Structure Interaction. Experience spans CFD, experimental aerodynamics, flight dynamics and numerical analysis, with computed results carried through to an independent check: NACA 0012 lift within 1.4% of NASA measurement, and a launch vehicle airbrake drag map verified against closed-form boundary layer theory to within 13%.

EDUCATION

Imperial College London

London, United Kingdom | Oct 2026 – Oct 2027

MSc ADVANCED COMPUTATIONAL METHODS FOR AERONAUTICS, FLOW MANAGEMENT AND FLUID-STRUCTURE INTERACTION

- Incoming student, October 2026. Core computational fluid dynamics and applications, advanced fluid mechanics and fluid-structure interaction, computational linear algebra; options in turbulence modelling, aeroelasticity and high-performance computing.

University of Bristol

Bristol, United Kingdom | Sep 2023 – Jul 2026

BEng AEROSPACE ENGINEERING, FIRST CLASS HONOURS

- Highest mark in the cohort in **Aerodynamics (95%)** and **Engineering Mathematics (94%)**; Structures and Materials 94%; Aerodynamics and Numerical Simulation Methods 77%.
- Further modules in flight dynamics and control, advanced structures and sustainable aircraft propulsion.

TECHNICAL SKILLS

Aerodynamics and CFD: ANSYS Fluent and CFX, Siemens STAR-CCM+, XFOil; steady RANS ($k-\omega$ SST), mesh and turbulence model selection, grid convergence and grid convergence index, y^+ , verification and validation.

Experimental Aerodynamics: Wind tunnel testing with a six-component balance, LabVIEW acquisition, CFD-to-test correlation, measurement uncertainty.

Flight Dynamics and Simulation: Nonlinear six-degree-of-freedom aircraft models, trim and linearisation, PI control and gain scheduling across flight conditions.

Programming and Numerical Analysis: Python, MATLAB, Simulink, C++; data reduction pipelines, automated testing of numerical code.

Structural and Multidisciplinary Tools: Abaqus, ANSYS Mechanical, one-way fluid-structure coupling, classical laminate theory, Autodesk Inventor.

FLIGHT PHYSICS EXPERIENCE AND PROJECTS

HyPower Bristol

Bristol, United Kingdom | Oct 2024 – Sep 2025

AERODYNAMICS AND STRUCTURES ENGINEER (PART-TIME)

- Ran a **25-case STAR-CCM+ sweep** of airbrake deployment against velocity for Project Waxwing, giving the brake controller a computed drag map instead of an assumed constant.
- Verified the CFD against closed-form theory before release, matching a Prandtl-Schlichting boundary layer prediction to within 13%, and reported the work as verified rather than validated because no measured data was set against it.
- Designed the wind tunnel validation campaign for the airbrake and fin flow interaction: scaled model, instrumentation and an eleven-configuration test matrix covering brake deployment and yaw offsets.
- Settled fin section and planform on evidence, selecting a hybrid NACA and wedge profile on CFD drag cross-checked against a MATLAB panel method, and a swept trapezoidal planform through a weighted decision matrix.

Aerodynamic CFD and Fluid-Structure Interaction

2026

INDEPENDENT STUDY IN ANSYS FLUENT AND MECHANICAL 2026 R1

- Solved 2D steady RANS flow over a NACA 0012 at 10° incidence and chord Reynolds number 6×10^6 , selecting mesh topology, turbulence model and convergence criteria.
- Validated against NASA measurement: computed C_p tracks experiment across the full chord and integrated lift gives $C_l = 1.06$ against 1.07 to 1.08 measured, 1.4% low, cross-checked independently against thin-aerofoil theory.
- Quantified discretisation error over a six-case one-factor-at-a-time matrix by Richardson extrapolation and the grid convergence index.
- Audited y^+ and near-wall treatment, identifying a mesh-model inconsistency over much of the chord and documenting its effect on solution fidelity.
- Solved a three-bladed horizontal-axis turbine in a rotating reference frame on a 120° periodic sector, mapping pressure and centrifugal load onto an orthotropic composite shell blade as a one-way coupling: 0.405 m tip deflection, root radial reaction within 0.116% of the analytic $m\omega^2 r$.

Experimental Aerodynamics, Blood-Transport UAV (AVDASI 2)

2024 – 2025

AERODYNAMICS LEAD AND PROJECT MANAGER, 60-STUDENT FULL-SCALE AIRCRAFT PROGRAMME

- Selected the NACA 2414 section for a full-scale 2.6 m UAV and owned the fairing drag decomposition, comparing XFOil predictions against wind tunnel measurement.
- Quantified experimental uncertainty on the fairing configuration: the 0.09 N measured drag delta fell below the 0.34 N run-to-run scatter, reported as statistically inconclusive rather than as a drag reduction.
- Delivered every structural joint and aerodynamic fairing on the aircraft, ten flight parts at 1,011 g, inside a 535-file Autodesk Inventor assembly, and led the fuselage design team through integration against the requirement set.

Pitch-Rate Control of the Stevens and Lewis F-16 Model

2025

FLIGHT DYNAMICS AND CONTROL STUDY, MATLAB AND SIMULINK

- Trimmed and linearised the full nonlinear six-degree-of-freedom aircraft model, verifying each linear model against the nonlinear plant at its operating point.
- Designed a PI pitch-rate controller and gain-scheduled it across two flight-envelope points, 220 and 100 m/s, holding closed-loop response consistent as the linearised dynamics changed.

STRUCTURAL AND RESEARCH EXPERIENCE

Bristol Composites Institute, University of Bristol

Bristol, United Kingdom | Jun 2025 – Aug 2025

RESEARCH INTERN, THROUGH-THICKNESS REINFORCEMENT OF COMPOSITES

- Planned and ran a test programme and its analysis with custom Python data reduction, roughly doubling Mode II delamination resistance with a new reinforcement architecture.
- Modelled the laminates in Abaqus using competing approaches, including cohesive zone fracture analysis, predicting interlaminar failure within about 10% of measurement.

Thin-Ply Quad and Double-Double Composite Laminates

2025 – 2026

FINAL-YEAR RESEARCH PROJECT (AENG30017), SUPERVISED BY PROF. MICHAEL R. WISNOM

- Manufactured and autoclave-cured four stiffness-equivalent 32-ply architectures from spread-tow prepreg, then tested them to failure on an Instron 1342 with video gauge strain measurement: 496 to 538 MPa at 1.41 to 1.53% strain.
- Built a ply-by-ply 3D Abaqus free edge model alongside classical laminate theory, tested the significance of the differences between architectures, and backed the pipeline with 65 automated tests.

AWARDS AND ACTIVITIES

- **Winner (Local) and Global Nominee, NASA International Space Apps Challenge**, 2025. Satellite-data land use suitability tool for Addis Ababa planners.
- **3rd Place, National Rocketry Championship**, 2024. Co-led the team that designed, built and launched a rocket carrying a Geiger counter payload measuring radiation with altitude.
- **Assembling and Testing a Quadcopter**, MathWorks verified certificate, 2024. Subsystem integration, sensors and data acquisition in MATLAB and Simulink.
- **Team Lead, Engineers Without Borders**, 2023. Water supply and purification scheme for Pu Ngaol, Cambodia, down-selected on maintainability. Event Organiser, Lebanese Society, 2023 to 2026; AeroSoc member throughout the BEng.